

**DUY TAN UNIVERSITY
INTERNATIONAL SCHOOL**



GROUP PROJECT

SUBJECT: CDIO

**PROJECT ELECTRONICS PRICE
COMPARISON WEBSITE**

INSTRUCTOR : NGUYEN DUC MAN

TEAM MEMBER : NGUYEN VAN DUY MANH

TRAN VIET LOC

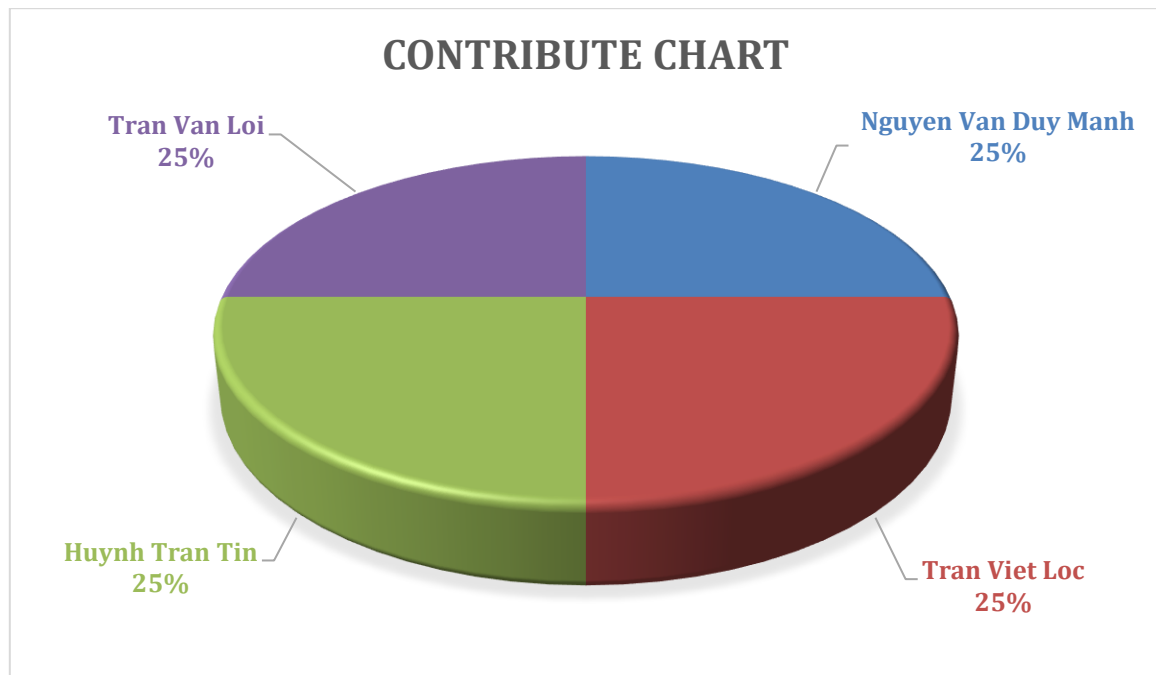
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TRAN VAN LOI

Da Nang, April 11, 2026

CONTRIBUTION TABLE

Members	Role	% Contribute
Nguyen Van Duy Manh	Backend + AI + Architecture	25%
Tran Viet Loc	Scrapping + Data	25%
Huynh Tran Tin	QA + Bug Fix	25%
Tran Van Loi	UI/UX + Frontend	25%



Link Github: <https://github.com/NV-DuyManh/Project-CDIO.git>

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I. Project Charter

1.1. Project Goal

Project Goal
<i>SMART: Specific, Measurable, Attainable, Relevant, and Time-bound</i> ● Successfully deploy a web application with functions such as Register, Login, Search Product, Compare Prices, Add to Favorites, Review & Comment, Change Password within 4 months of project launch. ● Build a database containing at least 1000 electronic products collected and continuously updated from e-commerce platforms. Achieve > 90% accuracy in product matching and price comparison across multiple platforms. ● Achieve > 90% accuracy in displaying correct product information and price comparison. ● Complete all documentation including reports, user guides, and source code before the end of the course.

1.2. Deliverables

Deliverables
1. Complete Web Application: User and admin interfaces with all core functionalities (search, compare, review, favorites). 2. Database: MySQL/PostgreSQL database containing product information, prices, users, and reviews. 3. Data Crawling/API Integration: System to automatically collect product data from sources (Shopee, Lazada, Tiki...). 4. Project Documentation: Technical report, user guide, installation guide, presentation slides, and source code.

1.3. Business Case / Background

Business Case / Background
Why are we doing this? ● Prices of electronic products differ across platforms like Shopee, Lazada, and Tiki,

making it hard for users to know where to buy at the best price.

- Users currently have to open multiple websites and compare manually, which is time-consuming and inefficient.
- A centralized system can aggregate product data, provide quick comparisons, and help users make faster and more cost-effective purchasing decisions.

1.4. Benefits, Costs, and Budget

Benefits, Costs, and Budget

Benefits:

- For Users: Quickly compare prices across multiple platforms in one place, helping them choose the best deal and avoid overpaying.
- For Businesses: Increase product visibility and attract more potential customers through price transparency.

For Development Team: Gain practical experience in building scalable web systems, data crawling, and real-world deployment.

Costs:

- Development Cost: Time and effort of team members for designing, coding, testing, and deployment.
- Infrastructure Cost: Expenses for cloud hosting, database servers, and system deployment.
- Data Collection Cost: API subscriptions, crawling tools, and proxy services to gather product data.
- Maintenance Cost: Ongoing system updates, bug fixing, and performance optimization.

Budget needed:

- Estimated: ~10,000 USD
- Development: ~6,000 USD
- Infrastructure & Hosting: ~1,500 USD
- Data Crawling / API: ~1,500 USD
- Miscellaneous (domain, testing, deployment): ~1,000 USD

1.5. Scope and Exclusion

Scope and Exclusion
In-Scope: ● Develop a web application with core features: ○ Product search and filtering across multiple platforms ○ Price comparison between different e-commerce websites ○ Product detail view (price, description, source link) ○ User authentication (Register, Login, Change Password) ○ Reviews and comments from users ● Integrate data collection through APIs or web crawling ● Implement background data updating and caching mechanisms to improve performance ● Support image-based product search using AI recognition ● Build a responsive and user-friendly interface ● Ensure system performance with fast response time (≤ 3 seconds) Out-of-Scope: ● Mobile application development (IOS/Android) ● Direct online payment or checkout integration ● Full real-time synchronization with all e-commerce platforms ● Guarantee 100% accuracy or real-time updates of product prices ● Advanced AI-based recommendation system ● Integration with external enterprise systems (ERP, CRM, etc.)

1.6. Project Team

Project Team
Project Sponsor: Nguyen Duc Man Product Owner: Nguyen Van Duy Manh Scrum Master: Tran Van Loi Development Team: Nguyen Van Duy Manh (PO, Support), Huynh Tran Tin, Tran Viet Loc, Tran Van Loi (Scrum Master). Additional Stakeholders: End-users, Online shoppers, E-commerce platforms.

1.7. Measuring Success

Measuring Success	
What is acceptable:	
1. The web application runs stably and returns search results within ≤ 3 seconds 2. Price comparison information is clear, accurate, and easy to understand. 3. Users can easily search, compare, and interact (review, comment, favorite) without confusion 4. The system can handle at least 1000+ products with stable performance. 5. Core features (search, compare, authentication, review) work correctly without critical errors 6. All documentation, source code, and reports are completed and submitted on time.	

II. Requirements and Analysis

2.1. Requirement List for Electronics Price Comparison Website

2.1.1. Functional Requirements

ID	Requirement Name	Description
FR1	Product Search	The system shall allow users to search for products by entering keywords (e.g., iPhone, Samsung)
FR2	Price Comparison	The system shall automatically fetch and compare product prices from multiple e-commerce platforms.
FR3	Image-based Search	The system shall allow users to upload a product image and automatically detect and search for related products.
FR4	View Search Results	The system shall display a list of products including name, image, price, and source platform.
FR5	Data Crawling History	The system shall store and display previously crawled product data for users to review.
FR6	Filter & Search History	The system shall allow users to filter and search within stored product history by keyword or platform.
FR7	View Product Detail	The system shall allow users to view detailed product information via external links.

FR8	Add to Cart	The system shall allow users to add selected products to a shopping cart.
FR9	User Registration	The system shall allow new users to create an account using username, email, and password
FR10	User Login	The system shall allow registered users to log in using their credentials.
FR11	Session Management	The system shall maintain user login sessions securely after authentication.
FR12	Navigation	The system shall provide navigation between pages such as Home, History, Login, and Register.

2.1.2 Non-Functional Requirements

ID	Requirement Name	Description
NFR1	Performance	The system shall return search results within 3–5 seconds after user input.
NFR2	Scalability	The system shall support multiple concurrent users and large volumes of product data.
NFR3	Security	The system shall encrypt user passwords and protect authentication data.
NFR4	Usability	The system shall provide a clean, modern UI that is easy to use for first-time users.
NFR5	Availability	The system shall be available 24/7 with minimal downtime.
NFR6	Reliability	The system shall ensure accurate and consistent product price data from sources.
NFR7	Compatibility	The system shall work across modern browsers (Chrome, Edge, Firefox).
NFR8	Maintainability	The system shall be modular to allow easy updates and maintenance.

NFR9	Data Integrity	The system shall ensure that stored crawl data is not lost or corrupted.
NFR10	AI Accuracy	The system shall provide reasonably accurate product recognition from uploaded images.

2.2. System Workflow:

The system operates according to the following process:

1. User accesses the website:

The user opens the PriceHunt web application via a browser.

2. Enter product search keyword:

The user inputs a keyword (e.g., product name) into the search bar or uploads a product image.

3. System processes the request and collects data:

The system analyzes the input, performs data crawling from multiple e-commerce platforms, and aggregates relevant product information.

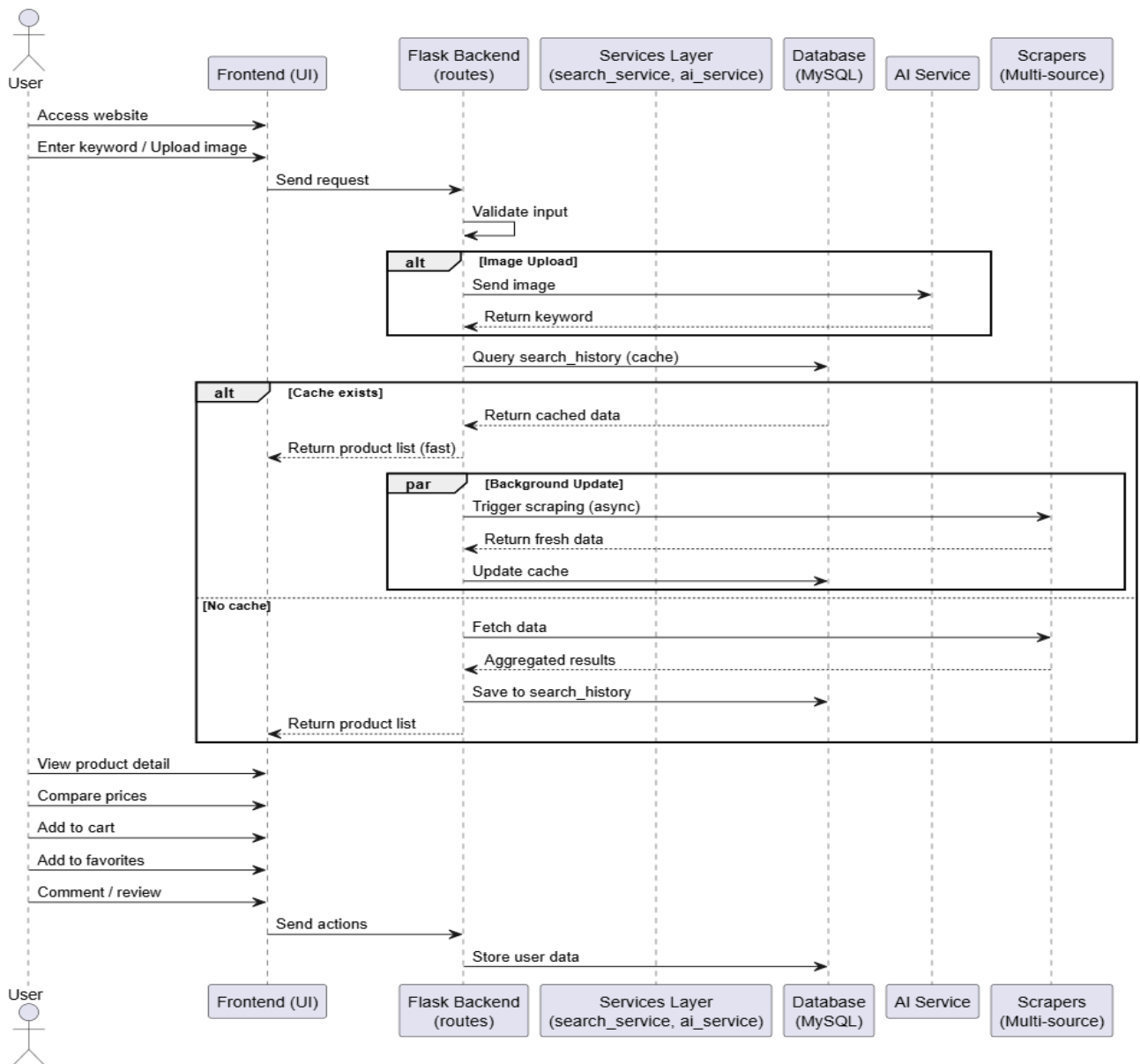
4. Display product list:

The system presents a list of products, including images, names, prices, and sources.

5. User interactions:

The user can perform the following actions:

- View product details: Access detailed information about a selected product.
- Compare prices: Compare prices of the same product across different platforms.
- View or add reviews: Read existing reviews or submit new reviews for products.



2.3. Database Design

2.3.1. Users Table:

Field Name	Data Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Unique user identifier
username	VARCHAR(80)	UNIQUE, NOT NULL	Username of the user
email	VARCHAR(120)	UNIQUE, NOT NULL	User email address
password_hash	VARCHAR(255)	NOT NULL	Encrypted password
is_admin	TINYINT(1)	DEFAULT 0	Role (0: user, 1: admin)
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Account creation time

2.3.2. Favorites Table:

Field Name	Data Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Unique favorite ID
user_id	INT	FK → users(id), NOT NULL	Owner of the favorite item
title	TEXT		Product title
price_str	VARCHAR(100)		Product price (string format)
img	TEXT		Product image URL
link	TEXT		Product link
site	VARCHAR(50)		Source platform (Shopee, Lazada, etc.)
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Time added to favorites

2.3.3. Cart Table:

Field Name	Data Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Cart item ID
user_id	INT	FK → users(id), NOT NULL	Owner of the cart
title	TEXT		Product title
price_str	VARCHAR(100)		Product price
img	TEXT		Product image
link	TEXT		Product link
site	VARCHAR(50)		Source platform
quantity	INT	DEFAULT 1	Quantity of product

2.3.4. Orders Table:

Field Name	Data Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Order ID
user_id	INT	FK → users(id), NOT NULL	User who placed the order

product_name	TEXT		Name of product
price	VARCHAR(200)		Product price
payment_method	VARCHAR(50)		Payment method (e.g., COD, Banking)
fullname	VARCHAR(100)		Customer full name
phone	VARCHAR(20)		Phone number
email	VARCHAR(120)		Email address
address	TEXT		Delivery address
status	VARCHAR(20)	DEFAULT 'paid'	Order status
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Order time

2.3.5. Comments Table

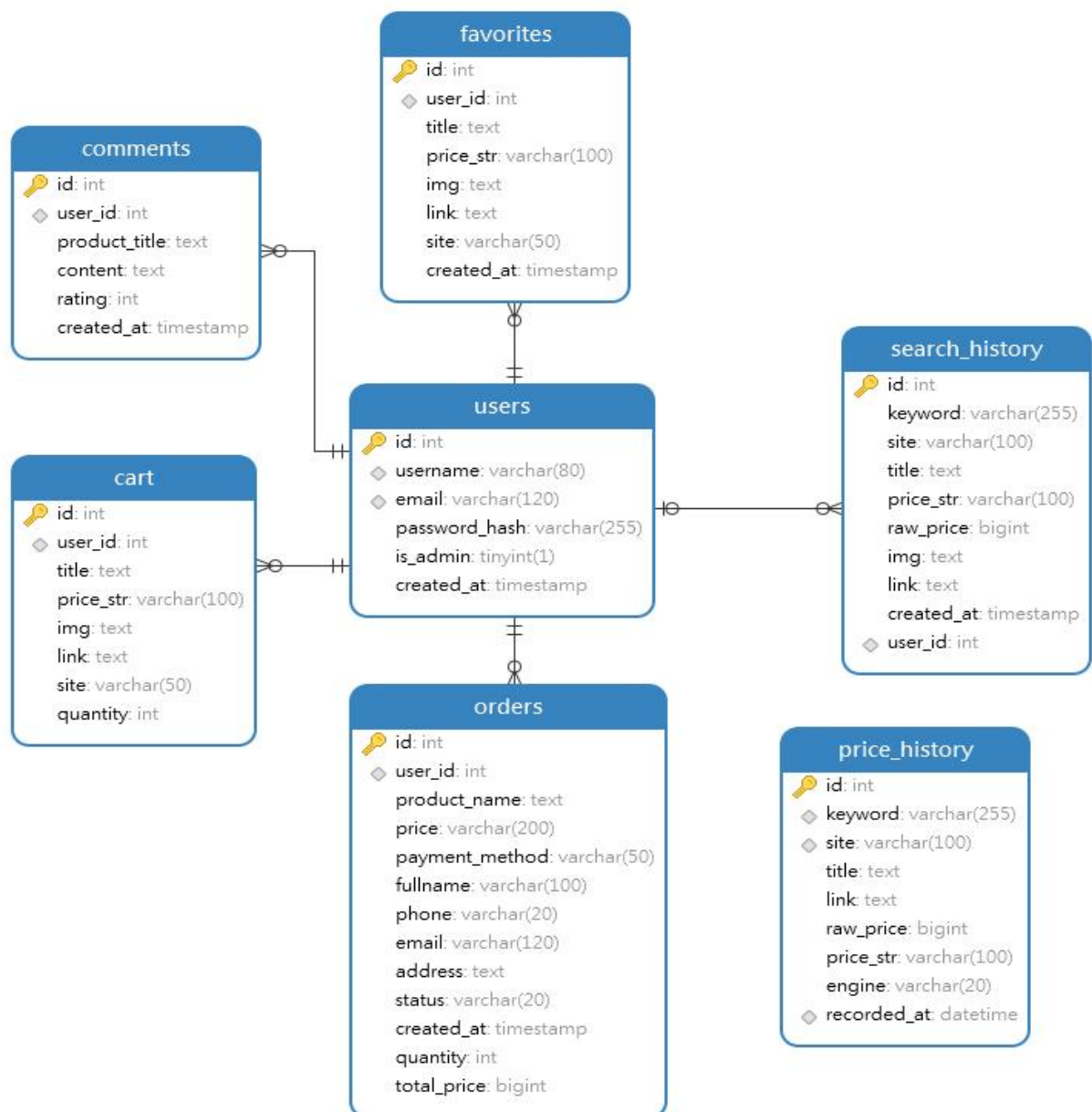
Field Name	Data Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Comment ID
user_id	INT	FK → users(id), NOT NULL	User who commented
product_title	TEXT		Product name
content	TEXT		Comment content
rating	INT	DEFAULT 5	Rating (1–5 stars)
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Comment time

2.3.6. Search History Table

Field Name	Data Type	Constraints	Description
id	INT	PK, AUTO_INCREMENT	Unique history record ID
user_id	INT	FK → users(id), NULL	User who performed the search (nullable if guest)
keyword	VARCHAR(255)	NOT NULL	Search keyword entered by user

site	VARCHAR(50)		Source platform (Shopee, Lazada, etc.)
title	TEXT		Product title
price	VARCHAR(100)		Product price (string format)
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Time of search

2.3.7. ERD



2.4. Technology Stack:

2.4.1. Frontend (HTML, CSS, JavaScript):

- **Authentication Module:**
 - Register, Login, Logout
 - Session-based authentication
 - Basic user validation (email, password)
- **Product Module:**
 - Search products by keyword
 - Display product list (title, price, image, source)
 - View product details via external links
- **Comparison Module:**
 - Display product prices from multiple platforms
 - Allow users to visually compare prices
- **User Interaction Module:**
 - Add/remove products to/from favorites
 - Add products to cart
 - Write reviews and comments
 - View history of crawled products
- **UI/UX Module:**
 - Responsive design (desktop-focused)
 - Simple and user-friendly interface
 - Loading indicators when fetching data
 - Error handling (no results, failed crawl, etc.)

2.4.2. Backend (Python Flask)

- **Authentication & Authorization:**
 - Session-based authentication (Flask session)
 - Password hashing (Werkzeug/bcrypt)
 - Basic role management (user/admin)
- **Product Processing:**
 - Handle search requests from users
 - Process keyword or image input
 - Handle search requests from users

- **Price Comparison Engine:**
 - Aggregate product prices from multiple e-commerce sources
 - Normalize data (price format, product name)
 - Return comparable results
- **Data Crawling Module:**
 - Web scraping using BeautifulSoup and Requests
 - Extract product information (title, price, image, link)
 - On-demand crawling when user searches
- **User Interaction Handling:**
 - Manage favorites, cart, and orders
 - Store and retrieve user comments/reviews

2.4.3. Database (MySQL)

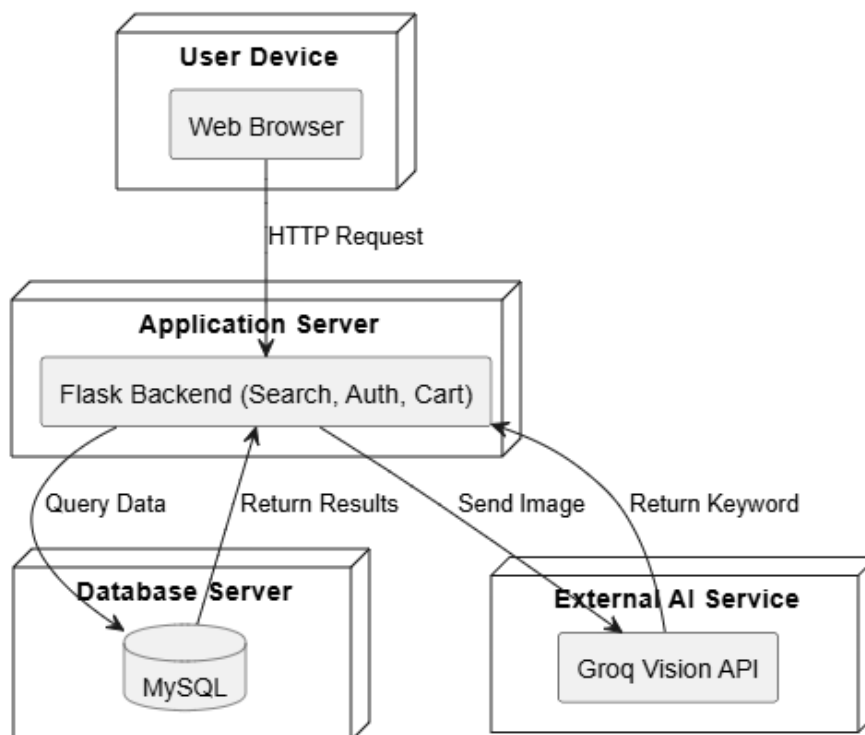
- **Database Management:**
 - Users
 - Use relational schema with foreign keys
- **Main Tables:**
 - Users
 - Favorites
 - Cart
 - Orders
 - Comments
- **Optimization:**
 - Indexed user_id for faster queries
 - Use InnoDB for transaction support
- **Additional Tools:**
 - Git: Version control
 - Browser: Chrome/Edge for testing
 - Deployment: Localhost / simple hosting

2.5. Project Schedule (Gantt Overview)

- **Week 1 – Requirement Analysis**
 - Activities: Gather requirements, define scope, create use cases
 - Members: Entire team

- **Week 2–3 – System Design**
 - Activities: UI/UX design, database schema, architecture planning
 - Members: Entire team
- **Week 4–6 – Front-end & Back-end Development**
 - Activities: Implement all modules (Auth, Tracking, Calendar, Insights, Admin)
 - Members: Entire team
- **Week 7–8 – Module Testing & System Integration**
 - Activities: Unit tests, module-level testing, FE–BE integration
 - Members: Entire team
- **Week 9 – Deployment & Stabilization**
 - Activities: Build release, deployment, bug fixing, performance adjustments
 - Members: Entire team
- **Week 10 – Security, Backup & Final Review**
 - Activities: Security enhancements, backup/recovery setup, final checks
 - Members: Entire team

III. UI/UX



IV. Experimental results

- **Performance:**
 - First search (no cache): 5-7 seconds
 - Cached search: 3-5 seconds
- **Accuracy:**
 - Price comparison accuracy: ~90%
 - AI image recognition accuracy: ~85%
- **Data:**
 - Database contains 1000+ products
 - Data collected from multiple platforms
- **Conclusion:**
 - System meets performance and usability requirements

V. Demo



PriceHunt

Tìm Kiếm

Lịch Sử

Giỏ hàng

0

Đơn hàng

duymanh

Đăng xuất

Kết quả cho "iPhone 15"

56 sản phẩm

Cache

Hiện thị

Tất cả

Clickbuy

CellphoneS

Di Động 3A

Smart Việt

Bách Long

Tiến Trần

CLICKBUY

Giá Tốt Nhất

iPhone 15 Lock 128GB

9,990,000 đ

Tối nơi bán

CLICKBUY

iPhone 15 128GB cũ đẹp 99%

11,990,000 đ

Tối nơi bán

CLICKBUY

iPhone 15 Plus Lock 128GB

11,990,000 đ

Tối nơi bán

CLICKBUY

iPhone 15 Plus 128GB Cũ đẹp 99%
- Không zin tặng máy

13,690,000 đ

Tối nơi bán

CLICKBUY

iPhone 15 256GB cũ đẹp 99%

14,390,000 đ

Tối nơi bán

CLICKBUY

CLICKBUY

BÁCH LONG STORE

SMART VIỆT

SMART VIỆT

PriceHunt

Tìm Kiếm

Lịch Sử

Giỏ hàng

0

Đơn hàng

duymanh

Đăng xuất

Đơn hàng của bạn

Theo dõi toàn bộ lịch sử mua hàng

MÃ ĐƠN	SẢN PHẨM	SL	ĐƠN GIÁ	THÀNH TIỀN	PHƯƠNG THỨC TT	TRẠNG THÁI	NGÀY ĐẶT
#42	Samsung Galaxy S23 Ultra (5G) Cũ Đ...	x3	11,790,000 đ	35,370.000đ	credit_card	✓ Đã thanh toán	2026-04-02 21:38:32
#43	Samsung Galaxy S23 Ultra 8GB 256G...	x1	12,090.000đ	12,090.000đ	credit_card	✓ Đã thanh toán	2026-04-02 21:38:32
#28	Apple iPhone 15 Pro New Bản VN/A - ...	x1	24,400.000đ	24,400.000đ	credit_card	✓ Đã thanh toán	2026-03-10 09:04:33
#29	Xiaomi Redmi Note 14 6GB 128GB	x2	4,290.000đ	8,580.000đ	credit_card	✓ Đã thanh toán	2026-03-10 09:04:33
#30	Xiaomi Redmi Note 13 Pro 8GB 256GB	x2	4,990.000đ	9,980.000đ	credit_card	✓ Đã thanh toán	2026-03-10 09:04:33
#31	Xiaomi Redmi Note 15 6GB 128GB	x4	5,690.000đ	22,760.000đ	credit_card	✓ Đã thanh toán	2026-03-10 09:04:33
#32	Xiaomi Redmi Note 14 5G 8GB 256GB	x2	6,490.000đ	12,980.000đ	credit_card	✓ Đã thanh toán	2026-03-10 09:04:33

PriceHunt

Tìm Kiếm

Lịch Sử

Giỏ hàng

0

Đơn hàng

duymanh

Đăng xuất

Lịch Sử Cào Dữ Liệu

Toàn bộ sản phẩm đã được lưu vào database

+ Tìm kiếm mới

Lọc theo tên, từ khóa, sản...

Tất cả

Clickbuy

Di Động 3A

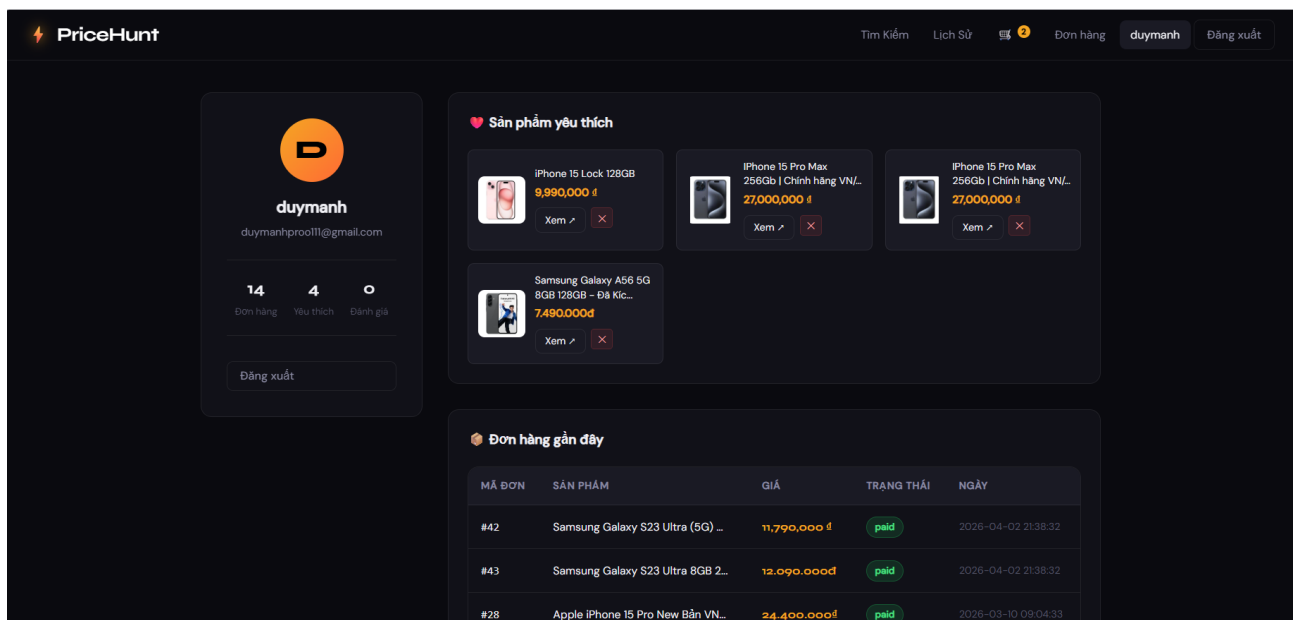
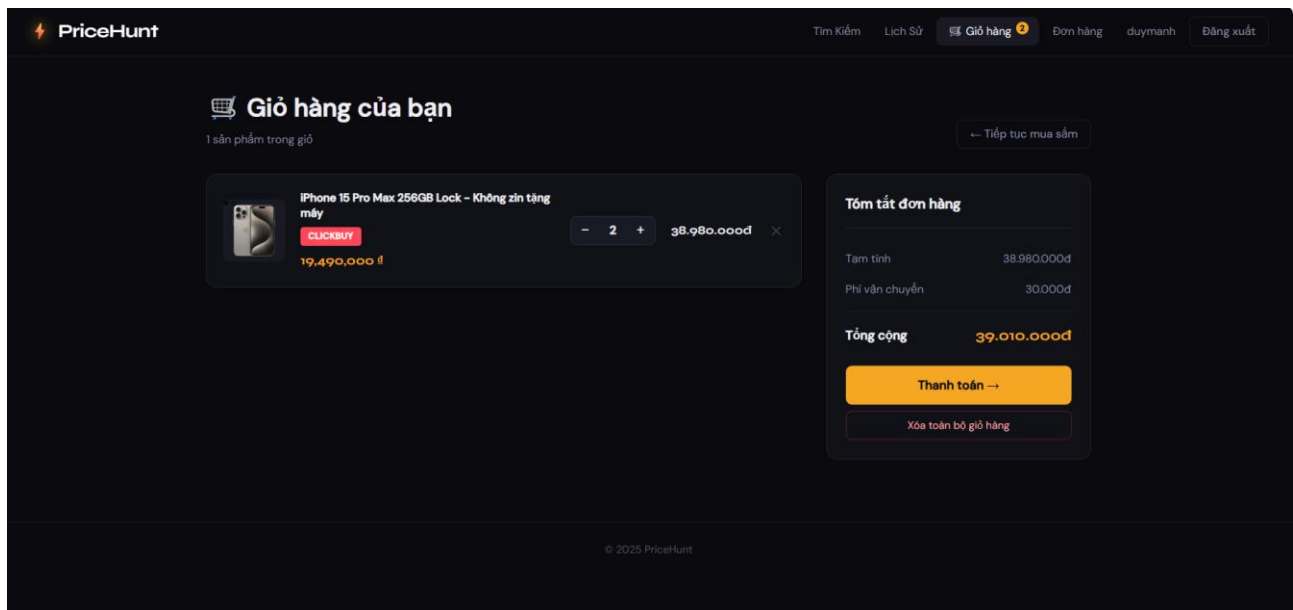
CellphoneS

Bách Long Store

Tiến Trần Mobile

Smart Việt

TỪ KHÓA	SẢN	HÌNH ẢNH	TÊN SẢN PHẨM	GIÁ BÁN	THAO TÁC	THỜI GIAN
iPhone 15	CLICKBUY		iPhone 15 Pro Max 256GB Cũ Đẹp - Không Zin...	18,290,000 đ	Xem Thêm vào giỏ hàng	2026-04-11 17:35:19
iPhone 15	DI ĐỘNG 3A		iPhone 15 Pro Max (1TB) Chính Hãng VN/A	40,590.000đ	Xem Thêm vào giỏ hàng	2026-04-11 17:35:19
iPhone 15	DI ĐỘNG 3A		iPhone 15 Pro Max (1TB) Quốc Tế Sách Tay	36,990.000đ	Xem Thêm vào giỏ hàng	2026-04-11 17:35:19
iPhone 15	CELLPHONES		iPhone 15 Pro 1TB Chính hãng VN/A	35,990.000đ	Xem Thêm vào giỏ hàng	2026-04-11 17:35:19
iPhone 15	DI ĐỘNG 3A		iPhone 15 Pro Max (512GB) Chính Hãng VN/A	35,390.000đ	Xem Thêm vào giỏ hàng	2026-04-11 17:35:19
iPhone 15	BÁCH LONG STORE		Apple iPhone 15 Pro Max 1TB (Nhập khẩu)	32,950.000 đ	Xem Thêm vào giỏ hàng	2026-04-11 17:35:19



Demo Detail Project:

https://drive.google.com/drive/folders/1oJB6d2f7QZySVCodPlgNeBRqNnb5q1S?usp=drive_link